

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance charges.

1.) Identify your problem statement

As a data scientist, we need to **predict the insurance charges based on age, sex, BMI, Noofchildren, Smoking habit.**

Domain Selection-Machine Learning

Learning Selection-Requirement is clear and input and output dataset is there ,hence it is **Supervised Learning** and **Regression**

2.) Tell basic info about the dataset (Total number of rows, columns)

Dataset having below details

Input-age,sex,BMI,Noofchildren,Smokinghabits

Output- Insurance Charges

No of rows- 1338

No of Columns-6

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Used pre-processing method for columns **Sex and Smoker**

Used Nominal method for preprocessing because both columns don't have any ordinal information.

4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Created Multiple Linear Regression, Support Vector Machine, Decision Tree and Random Forest

5.) All the research values (r2_score of the models) should be documented.

(You can make tabulation or screenshot of the results.)

Multiple Linear Algorithm

Standardisation	R2 score
Without Standardizations	0.7894790349867009
After Standardization	0.789479034986701

Support Vector Machine

Kernel	C value	R2 value
Default	Default	-0.08338238593619329
sigmoid	10	0.03930714378274347
sigmoid	100	0.5276103546510407
sigmoid	1000	0.28747069486976173
rbf	10000	0.8779952401449918
rbf	20000	0.8780247463619089
linear	100	0.8780247463619089
poly	1000	0.8566487675946572
linear	1000	0.7649311738597411

Decision Tree

critierion	splitter	R2 value
squared_error	best	0.6873182829498646
absolute_error	best	0.6947337589493879
poisson	best	0.7335577979685567
squared_error	random	0.70780549131693
absolute_error	random	0.7184883002751841
poisson	random	0.6697491395444315

Random Forest

critierion	n_estimators	R2 value
squared_error	100	0.8554815574427125
squared_error	10	0.8398833836428838
squared_error	1000	0.8542881070181388
absolute_error	300	0.8524812760203333
absolute_error	400	0.854723379457845
poisson	100	0.8544595170133349
poisson	1000	0.8576276734031387

6.) **Mention your final model, justify why u have chosen the same.**

Chosen **Support Vector machine** Algorithm as final model with parameter

Kernel=Linear and C=100